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Electrochemical Method Of Analysis

"The analytical techniques that involved the measurement of electrical properties of a system to obtain information about the composition or concentration of chemical species are called **electrochemical methods of analysis**"

- These methods are based on the principle using an electrochemical cell, which consists of an electrode (working electrode) and a reference electrode immersed in an electrolyte solution containing the analyte of interest.

Several electrochemical methods are used for the analysis including

- i) Potentiometry
- ii) Voltammetry
- iii) Amperometry
- iv) Coulometry
- v) Impedance Spectroscopy
- vi) Conductometry

Conductometry

The branch of electrochemistry which is based on the measurement of conductance of the electrolyte is called **conductometry**.

- It is based on the principle that when electric potential is applied across an analyte, ions in the solution move under the influence of electric field, resulting in the flow of electric current.

To a view point of electrical conduction, substances are divided into two categories

(1) Insulators

(2) conductors

conductors are further subdivided into three main types

- i) Metallic or electronic conductors
- ii) Electrolytic or ionic conductors
- iii) Mixed conductors

Types of Conductometry:

There are two types of conductometry on the basis of nature of measurement i.e.

- 1) Direct Conductometry
- 2) Indirect Conductometry

Indirect conductometry subdivided into three types.

- a) Titrimetric Conductometry
- b) Potentiometric Conductometry
- c) Polarographic Conductometry

Basic Principle of Conductometry:

When electric potential is applied across an analyte, ions in the solution move under the influence of electric field, resulting in the flow of electric current.

The movement of is mainly depends upon the conc. of ions.

- The electric conductance is in accordance with **Ohm's Law** which is state that:

"The strength of current (I) passing through conductors is directly proportional to potential difference (V)," as

$$I \propto V \Rightarrow I = RV \text{ or } V = IR, \text{ then}$$

$$R = \frac{V}{I} \quad (\text{ohm}(\Omega))$$

Some important definitions and relations:

Resistance:

"Hindrance in the flow of current is called **resistance**"

- It is represented by (R) and measures in **ohm (Ω)**
- It is calculated by following way:

According to ohm's law

$$I \propto V \Rightarrow V = IR \text{ and}$$

$$R = \frac{V}{I}$$

Conductance:

"Reciprocal of resistance is called **conductance**"

Conductance is mainly the measure of how easily an electric current flows through a conductor or a solution.

- It is denoted by (G) and its units are **ohm⁻¹ (Ω^{-1})/Siemens (S)**
- It is calculated as:

$$\text{Conductance } (G) = \frac{1}{\text{Resistance } (R)}$$

Specific Resistance/Resistivity:

"Resistance of a conductor of uniform cross section is directly proportional to its length and inversely proportional to its area of cross section is called **specific Resistance/Resistivity**"

Mathematically,

$$R \propto \frac{l}{A} \Rightarrow R = \rho \frac{l}{A} \Rightarrow \rho = \frac{RA}{l}$$

where ' ρ ' is a constant

called specific resistance or resistivity and can also be defined as:

"Resistance of a conductor of unit length and unit area of cross section is called **specific resistance or resistivity**"

- Material with high specific resistance are generally poor conductor of electricity and vice versa.
- It measures in ohm.cm / ohm.m

Specific Conductance / Conductivity:

"Reciprocal of specific resistance is called **specific conductance or conductivity**"

- It is represented by **K** (κ) and calculated as:

$$K = \frac{1}{\rho}$$

Its units are $\text{ohm}^{-1}\text{cm}^{-1} / \text{mhos}.\text{cm}^{-1}$

In case of a solution of electrolyte:

$$R \propto l \text{ --- ① and } R \propto \frac{1}{A} \text{ --- ②}$$

By combining ① and ② we get;

$$R \propto \frac{l}{A} \Rightarrow R = \rho \frac{l}{A} \text{ and } G = \frac{1}{R} = \frac{A}{\rho l},$$

$$G = \frac{1}{\rho} \frac{A}{l} \Rightarrow G = K \frac{A}{l} \text{ and } K = \frac{Gl}{A}$$

Where $\frac{l}{A}$ is called cell constant and define as:

"The distance b/w two parallel electrodes of a cell divided by area of cross section of electrodes is called **cell constant**"

$$\text{Cell constant (C)} = \frac{l}{A}$$

Equivalent Conductance:

"The product of specific conductance (K) and volume (V) of solution which contain one gram equivalent of electrolyte is called **equivalent conductance**"

- It is represented by ' λ_{eq} ' and calculated as $\lambda_{eq} = K \times V$
- Its unit are $\text{ohm}^{-1}\text{cm}^2$ per gram equivalent.
- If conc. of soln. is C - gram equivalent per litre then

$$\lambda_{eq} = \frac{K \times 1000}{C}$$

Molar Conductance:

"The product of specific conductance (K) and volume (V) which contain one mole of electrolyte is called **molar conductance**"

- It is represented by λ_m or Λ and calculated as:

$$\lambda_m = \Lambda = K \times V \Rightarrow \lambda_m = \frac{K \times 1000}{M}$$

Its units are $\text{ohm}^{-1}\text{cm}^2$ per mole

Relation b/w λ_{eq} and λ_m :

$$\lambda_m = \frac{K \times 1000}{M} \text{ and } \lambda_{eq} = \frac{K \times 1000}{N}$$

We know that,

$$\text{Normality} = \text{Valency factor} \times \text{Molarity} \quad (N.F.)$$

$$N = m \times M$$

So,

$$\lambda_{eq} = \frac{\lambda_m}{V.F.}$$

Factors affecting the ^{specific} conductance of electrolyte:

a) Inter ionic attraction:

If inter ionic attraction b/w solute is more, conductivity will be less.

b) Polarity of solvent:

If the solvent is greater polarized, then the ionization and conductivity will be more.

c) Viscosity of medium:

By increasing the viscosity of medium, conductivity decreases.

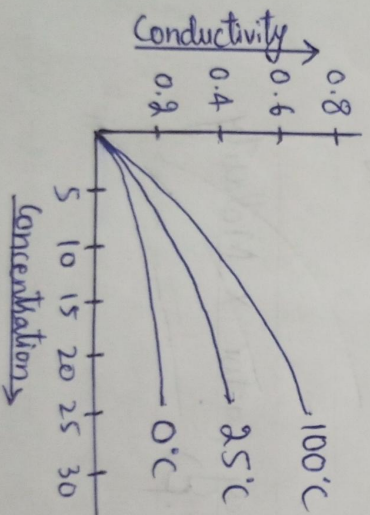
d) Temperature:

- As the temperature of electrolyte solution is increased, the conductivity increases because K.E of ions increases and types of attractive forces decreases and viscosity of medium decreases.

Example:

A family of curves given below, shows that the change in conductivity of sodium chloride with temperature.

- With increase in temperature, the conductivity is also increased due to high availability of ion carriers and with decrease in temperature, the conductivity also decreased.



e) Nature of electrolyte (Dilution):

For a strong electrolyte, the conductance decrease with dilution due to decrease in the number of current carrying

ions present per unit volume of the solution.

- For weak electrolytes, the value of conductance increase with dilution, as on dilution more ions of electrolyte ionizes.
- The degree of ionization increases with the increase of dilution of the solution, of the conductivity in increase due to increasing the number of ions.

Effect of dilution on specific conductance:

Specific conductance decreases with increase of dilution of the solution due to presence of no. of ions in 1 cm^3 solution decreases

Effect of dilution on Equivalent/Molar conductivity:

- The equivalent/molar conductivity increases with dilution. This is due to the fact that degree of ionization increases with dilution.
- When the whole of the electrolyte has ionized, further addition of the water bring a small change in the value of equivalent/molar conductance. This stage is called **infinite dilution**

- The ratio of equivalent conductivity at any dilution to equivalent conductivity at infinite dilution is called **conductivity ratio or degree of dissociation of solute.**

$$\alpha = \frac{\lambda}{\lambda_{\infty}}$$

How Conductance is measured?

Now consider an electrolyte 'MA' on electrolysis, produce M^+ and A^- ions. The M^+ ions moves towards cathode and gain e^- from cathode. A^- ions moves towards anode to lose their e^- . So net result is a flow of electrons through electrolyte or the electricity flows through the electrolyte.

The conductance of the electrolyte depends upon

the following two factors

- 1) Total number of elections present on cations and anions .
- 2) Velocity or Mobility of ions in the solution.

There are following factors which effect on the mobility or velocity of ions:

i) Nature of Solvent:

If we use polar solvent like water then speed becomes slow due to hydrogen bonding force.

If the solvent is non-polar then the mobility of ions is increased as compared to polar solvent

ii) Nature of ions:

The velocity of ions also depends upon the nature of ions e.g. Fe^{2+} and Fe^{3+} have different mobilities

a) charge on ion

b) No. of ions

iii) Size of ions:

Smaller ions will move faster but bigger ions will experience slow speed/mobility

iv) The applied Voltage:

The mobility of ions depends upon the voltage applied. If the voltage is less then ions will move slowly and vice versa

v) Viscosity and Temperature:

If the solution is viscous the mobility of ions will be less. Similarly, at low temperature mobility of ions again will be less

Kohlrausch's Law: This law is independent of ionic mobilities provides the basis for conductometric principle. It states that:

Conductance of solution is equal to the sum of conductances of individual ions present in solution

$$\Lambda = \Lambda(\text{cations}) + \Lambda(\text{anions}) \quad \text{e.g. } \Lambda_{\text{H}_2\text{SO}_4} = \Lambda(\text{OH}^-) + \Lambda(\text{H}^+)$$
$$= 198.6 + 349.8 = 548.4 \text{ mhos cm}^2 \text{ mole}^{-1}$$

The Measurement of Conductance : Instrumentation :

For the measurement of conductance

the following sources are required:

- | | |
|----------------------|----------------------|
| 1) Power Source | 2) Conductivity Cell |
| 3) Resistance Bridge | |
- ## 1) Power Source :

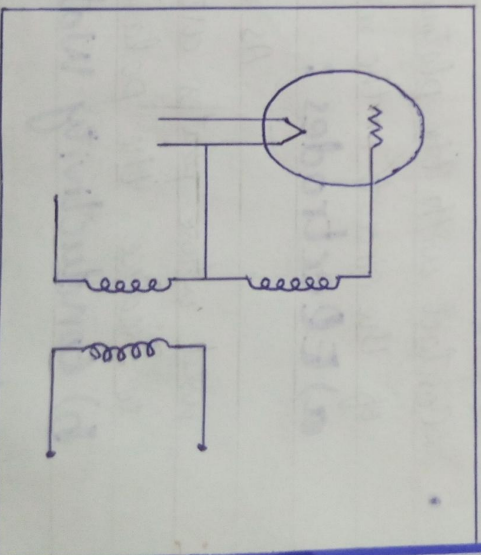
Alternating current of low intensity is used for all conductance measurements. It may have 50-10,000 Hz, however most commonly is 1000 Hz

- The A.C. source may be of a low voltage tapping on the 50 Hz transformer or transistor oscillator.
- A direct source current can't be used because it set up an E.M.F of polarization which tends to oppose the flow of current

- Hall and Adams suggested a vacuum tube oscillator for conductivity measurements in 1919.

- The circuit frequency oscillator is shown:

- The grid circuit of vacuum tube 'T' contains a grid coil 'L₁' and connected to the oscillator coil 'L₂' in parallel with variable condenser 'C'. The coil 'L₃' which is an output coil is coupled with 'L₂' inductively and conveys the oscillation to the conductance bridge.



- The 'vacuum tube oscillator' is better than other kinds of oscillators because it is silent in operation and produce alternating current of constant frequency.

2) Conductivity cell :

"The solution whose conductance is to be measured is placed in special type of cell known as

conductivity cell"

- Figure 'a' shows the type of cell employed for solution of low conductance solute, figure 'b' and 'c' are employed for highly conducting solutions.
- The electrodes consist of 'Pt' coated with finely platinum black and welded to platinum wires fused in glass tubes. The glass tubes contain mercury and are firmly fixed in the cover of cells so that the distance b/w the electrodes may not alter during the experiment.

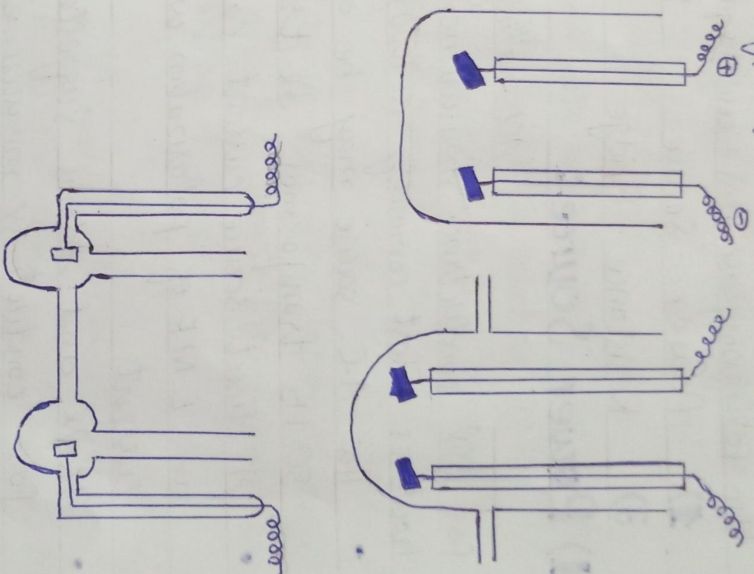
- Contact with the platinum is made by dipping the copper wires of the circuit in the mercury contained in the tubes

a) Electrodes :

As in the above cells, 'Pt' electrodes are used whose plates are coated with finely divided Pt black to reduce the polarization effect.

b) Conductivity water :

The conductivity water is used in the measurement of conductivity because it is sensitive for the presence of impurities and dissolved CO_2 . Distilled water is not used due to above mention reasons



3) Resistance Bridge:

The specific conductance of a solution can't be determined by finding the resistance of the solution by means of wheat stone Bridge arrangement.

- The main difficulty in determining the resistance of the solution of an electrolyte is that electrolysis of solution takes place simultaneously with the conduction of current and this produce two complications

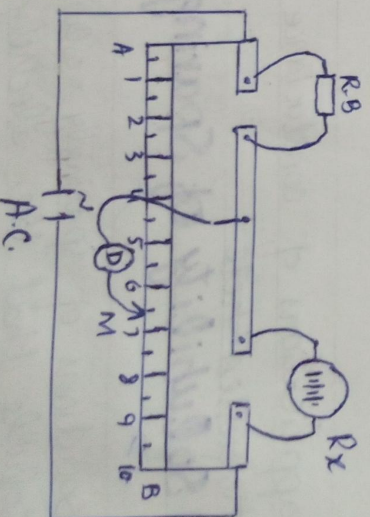
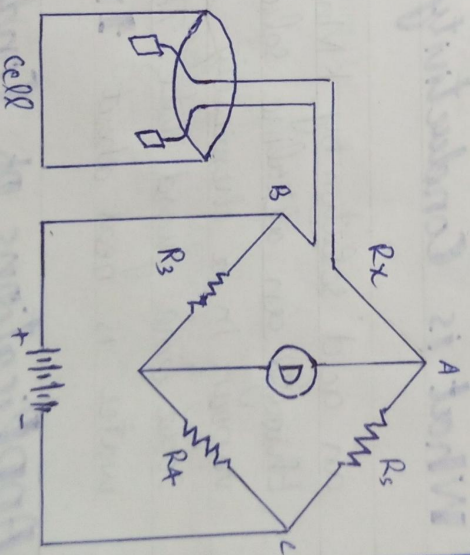
i) Polarization soon gets in and cause the resistance to vary.
ii) The concentration of the solution alters

- To avoid these complications, an alternating current is employed. Since under these conditions a galvanometer can't be used as in wheatstone bridge circuit, it is replaced by a head telephone or magic eye
- If there is no flow of current through the circuit as indicating by the magic eye or head phone, the following relationship exists.

$$\frac{R_x}{R_5} = \frac{R_3}{R_4}$$

- If the values of resistance of three conductors i.e. R_2, R_3 and R_4 are known, the value of unknown resistance ' R_x ' can be calculated. The apparatus shown in the figure.

- A suitable resistance is introduced by means of the resistance box and moving contact 'M' is moved along the wire 'AB' to a point where there is no flow of current as indicated by the head phone at this length AM and BM recorded.



$$\frac{R_x}{R_B} = \frac{BM}{AM}$$

- Knowing the resistance of resistance box and lengths AM and BM , the specific resistance ' K_s ' of solution placed in conductivity cell can be calculated. The reciprocal of specific resistance will give the conductance of the solution.

$$R_x = R_B \times \frac{BM}{AM}, \text{ observed conductance will be as}$$

$$\text{observed conductance} = \frac{1}{R_x} = \frac{1}{R_B} \times \frac{AM}{BM}$$

$$\text{As we know that, } R \propto \frac{l}{a} \Rightarrow R = \rho \frac{l}{a} \therefore \frac{l}{a} = x, \text{ and}$$

$$R = \rho \times x \text{ (where 'x' is called cell constant)}$$

$$x = \frac{R}{\rho} \Rightarrow x = \frac{1/\text{observed conductance}}{1/\text{specific conductance}}$$

$$\text{Specific conductance} = \text{observed conductance} \times x$$

From above equation

$$\text{Specific conductance} = \left(\frac{1}{R_B} \times \frac{AM}{BM} \right) \times \text{cell constant}$$

What is Conductivity Water :

It is obtained by distilling an acid solution by $KMnO_4$ in ordinary distilled water through an alkaline solution of $KMnO_4$ and condensing the vapours in a chemically resistance glass flask provided with a soda lime guard tube. The specific conductivity of such a water is near about 1×10^{-6} ohms

Applications of Conductometry:

There are many applications of conductance measured. Some of these are discussed here

1) Solubility of Sparingly Soluble Salts :

Saturated solutions of sparingly soluble salts is such that salt is so soluble that the electrolyte present in it may be regarded

as completely ionized. The relation b/w the equivalent conductance ' λ ' and specific conductance ' k ' and the concentration ' c ' is given by the equation:

$$\lambda_v = K_v \times \frac{1000}{c} \quad (\text{Where 'c' represents solubility in gram eq. per litre}).$$

Suppose, we have to find out the conductivity of AgCl at 25°C. The AgCl is first washed with conductivity water to remove all the soluble impurities and then AgCl is dissolved in conductivity water and warmed, then kept at 25°C. The solution is said to be saturated and solid AgCl will settle down. The conductivity of this solution can easily be determined. Since concentration of AgCl is present in the soln. The value of λ_v can be considered to be equal to λ_∞ i.e. Equivalent conductance at infinite dilution. Thus we can say,

$$\lambda_v = \lambda_\infty = K_v \times V \quad (\text{Where 'V' is volume of 1g eq. of solute.})$$

According to Kohlrausch's law:

$$\begin{aligned} \lambda_\infty &= \lambda_\infty(\text{cations}) + \lambda_\infty(\text{anions}) \\ \lambda_\infty &= 61.92 + 76.35 \\ \lambda_\infty &= 138.27 \text{ mho's cm}^2 \text{ eq}^{-1} \end{aligned}$$

So from the values of K_v , λ_∞ and ' V ' containing 1g equivalent can be determined and hence solubility ' S ' can be calculated:

$$\text{Solubility (S)} = \frac{1000}{V} \times \frac{c}{\text{gm/e}}$$

$$S = \frac{1000 \times E \times K_v}{\lambda_\infty}$$

$$S = \frac{1000 \times 143.5 \times K_v}{138.27}$$

Hence knowing ' K_v ', the solubility of sparingly soluble salts can be calculated.

2) Ionic Product of Water:

Water ionizes as,
$$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^- \therefore K = \frac{[\text{H}^+][\text{OH}^-]}{\text{H}_2\text{O}}$$

Since water ionizes to very small extent, conc. of water may be considered almost constant i.e.

$K_w = [\text{H}^+][\text{OH}^-]$, The product of ionic concentration of H^+ and OH^- ions is expressed in grams moles per litre is constant at a constant temperature.

The ionic product of water can be calculated by knowing the specific as well as equivalent conductance at infinite dilution ' λ_{∞} ' is expressed by the equation:

$$\lambda_{\infty} = \frac{K_w}{C} \times 1000 \quad (K_w = 5.54 \times 10^{-8} \text{ mhos/cm}^2 \text{ at } 25^\circ\text{C}).$$

The value C of equivalent conductance of the two ions of water at infinite dilution are:

$$\lambda_{\infty}(\text{H}^+) = 349.8 \text{ mhos cm}^2$$

$$\lambda_{\infty}(\text{OH}^-) = 198.6 \text{ mhos cm}^2$$

$$\lambda_{\infty}(\text{H}_2\text{O}) = 548.4 \text{ mhos cm}^2$$

By substituting the values of ' K_w ' and ' λ_{∞} ' in equation:

$$\lambda_{\infty} = \frac{K_w}{C} \times 1000$$

we get the values of $[\text{H}^+]$ and $[\text{OH}^-]$ conc., Thus,

$$[\text{H}^+] = \frac{5.54 \times 10^{-8} \times 1000}{548.4} = 1.01 \times 10^{-7} \text{ moles}$$

$$[\text{OH}^-] = \frac{5.54 \times 10^{-8} \times 1000}{548.4} = 1.01 \times 10^{-7} \text{ moles}$$

Hence, ionic product of water is

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$K_w = [1.01 \times 10^{-7}][1.01 \times 10^{-7}]$$

$$K_w = 1.02 \times 10^{-14}$$

3) Basicity of Organic Compounds:

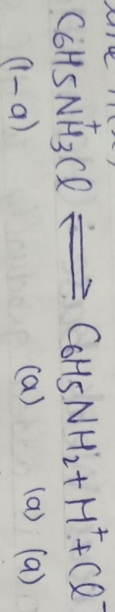
After conducting the studies on the sod. salts of a large no. of organic acids, Ostwald gave a empirical relation as:

$$B = \frac{\lambda_{1024} - \lambda_{32}}{10.8}$$

Where, λ_{1024} and λ_{32} are the conductance of two salts of strength $N/1024$ and $N/32$ respectively at 25°C and 'B' is the basicity of the acid.

4) Degree of Hydrolysis:

Aniline hydrochloride which is a weak base (Aniline) and a strong acid (HCl) when dissolved in water get hydrolyzed to produce equivalents of unhydrolyzed salts and equivalents of free acid and free base 'a' being the degree of hydrolysis of the salt (aniline HCl):



The hydrolysis constant (K_h) can be obtained by using the relation:

$$K_h = \frac{a^2 \cdot c}{1-a}$$

5) Degree of Dissociation of Weak electrolyte:

The degree of dissociation of weak electrolyte is given as:

$$\alpha = \frac{\lambda_v}{\lambda_{\infty}}, \text{ where } \lambda_v = \text{Eqv. conductance at given dilution}$$

λ_{∞} = Eqv. conductance at infinite dilution.

But the value of λ_{∞} can be calculated from equivalent conductance take at infinite dilution and ' λ_v ' can be calculated experimentally. Hence the degree of dissociation of weak electrolytes can be calculated.

6) Applications to Neutralization Titrations:

conductometric

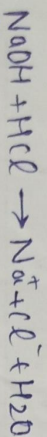
titration methods can be used very successfully for the determination of neutralization titrations.

Strong Acid Versus Strong Base : (HCl + NaOH)

On adding NaOH (st. base)

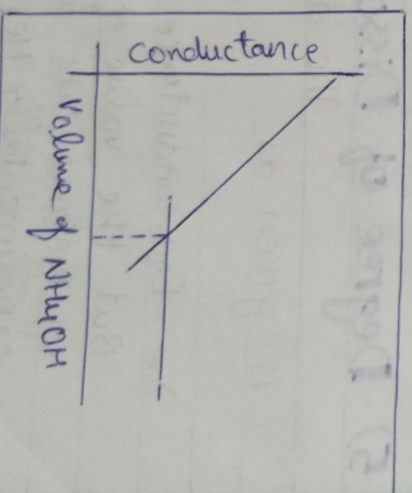
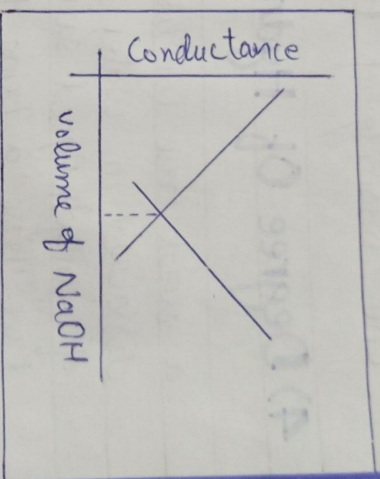
to HCl (st. acid) the more conducting H^+ ions of HCl are replaced by less conducting Na^+ ions, it means that H^+ ions decrease during titration. Hence conductivity falls upto the equivalence point. After the neutralization, the addition of more NaOH cause increase in the conductivity of soln. The graph obtain for such a titration has been shown in figure.

$$\lambda(H^+) = 349.8 \Omega^{-1}cm^2mol^{-1}, \lambda(Na^+) = 50.1 \Omega^{-1}cm^2mol^{-1}$$



Strong Acid Versus Weak Base : (HCl + NH_4OH)

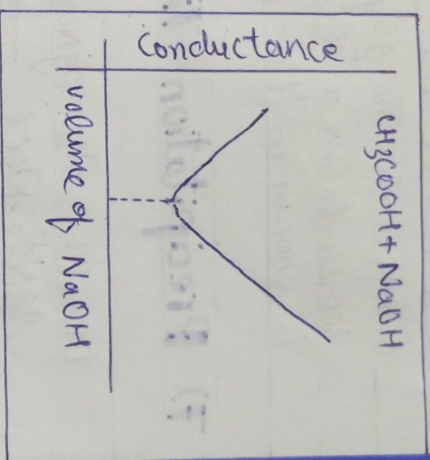
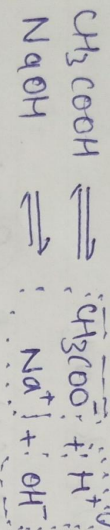
When NH_4OH is added gradually to HCl solution, there is replacement of highly conducting H^+ ions by less conducting NH_4^+ (73.4) ions so that conductivity of the titration liquid goes on decreasing. At the equivalence point all HCl is converted into NH_4Cl and further addition of NH_4OH does not cause any chemical change. Since NH_4OH is weak electrolyte its addition after the equivalence point will not cause any appreciable increase in conductance. The graph in this case is shown in the figure



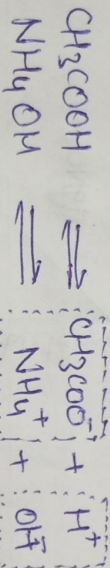
Weak Acid Versus Strong Base: ($\text{CH}_3\text{COOH} + \text{NaOH}$)

This titration gives an example of buffer whose pH remains constant and shows buffer action. The conductance decreases over this relatively short titration zone. The establishment of buffer soln stabilizes the $[\text{H}^+]$ to certain degree but with further addition of base up to the equivalence point, the conductance will gradually increase because of increase in $\text{Na}^+(\text{soln})$ and CH_3COO^- .

After the equivalence point a rapid increase in conductance occurs as the $[\text{OH}^-]$ increases, as shown in figure

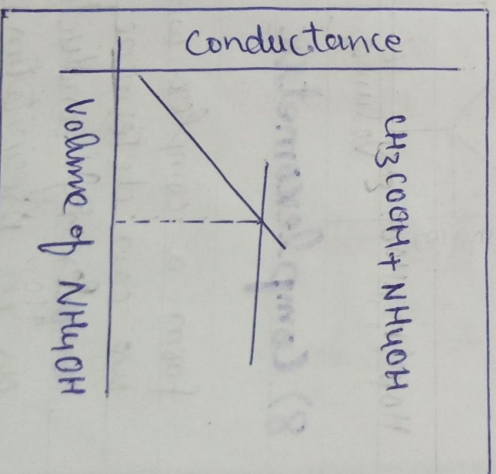


Weak Acid Versus Weak Base: ($\text{CH}_3\text{COOH} + \text{NH}_4\text{OH}$)



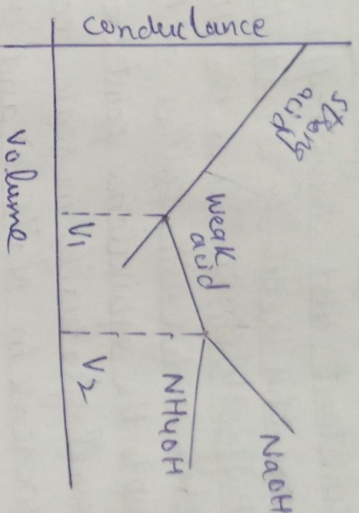
Since CH_3COOH is a weak acid

the conductivity of CH_3COOH soln taken for titration will be quite small on adding NH_4OH . There is a formation of $\text{CH}_3\text{COONH}_4$ which is a strong electrolyte so conductance will go on increasing until the equivalence point is reached. After the equivalence point the addition of NH_4OH would not cause any appreciable change in conductance of the titration liquid because it does not undergo any chemical change and is itself a weak electrolyte. Figure shows the type of titration curve obtained in such a case



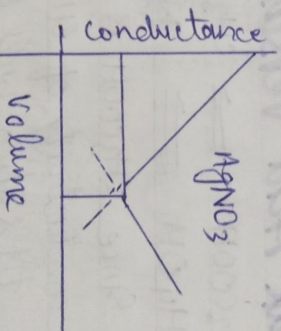
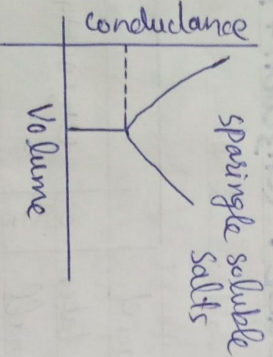
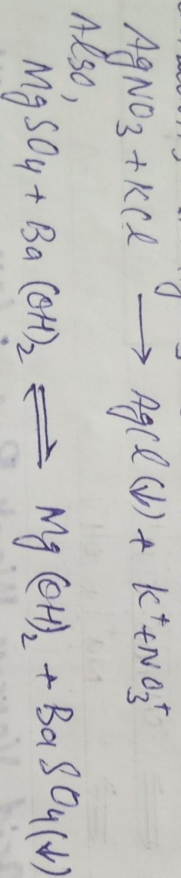
Mixture of a Strong acid and Weak base Versus Strong base:

A mixture of a strong acid and weak base can be used with a strong base. The graphical representation of this curve is shown in the figure.



7) Precipitation Titration:

As an example of a precipitation titration, 0.1M AgNO_3 can be titrated with 0.1M KCl solution



8) Complexometric Titration:

Any metal salt is being to form a complex with complex forming reagent (ligand), we can determine the amount of ligand which forms complex by conductometry. The curve will be same as in precipitation titration.

9) Determination of Vanilla in Vanilla essences:

Commercial vanilla essence contain about 1% vanilla though this conc. vary widely. By titration, conc of vanilla in essence can be calculated. At first known conc of essence is titrated to obtain the shape of curve, then unknown concentration can be calculated.